

Subject Description Form

Subject Code	EIE3112
Subject Title	Database System
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To introduce: 1. database design, development, and programming 2. advanced database queries and database security 3. NoSQL and cloud databases
Intended Subject Learning Outcomes	Upon completion of the subject, students will be able to: <u>Category A: Professional/academic knowledge and skills</u> 1. Database design, development, and programming 2. Advanced database queries and database security 3. NoSQL and cloud databases <u>Category B: Attributes for all-roundedness</u> 4. Communicate effectively
Subject Synopsis/ Indicative Syllabus	Syllabus: 1. <u>Database Design and Development</u> 1.1 DBMS systems; Client-server architecture; Database architectures and the web 1.2 SQL: data manipulation; data definition; 1.3 DB Development: DB applications and views; 1.4 Advanced SQL: SQL programming language; stored procedures; functions; triggers; cursors; exception handling 1.5 ER Modelling: ER diagrams; Transforming ER diagrams to relations 1.6 Normalization: Data redundancy and update anomalies; functional dependencies; normalization processes; normal forms 2. <u>Managing Database Environments</u> 2.1 Database Security: Database security best practices; SQL injection; Preventing SQL injection 3. <u>NoSQL and Cloud databases</u> 3.1 Introduction to NoSQL databases: overview, types, use cases and applications, basic operations, advantages and disadvantages 3.2 Introduction to Cloud databases: overview, types, use cases and popular cloud database platforms, key features, advantages and disadvantages

	Laboratory Experiments Lab 1: Database Implementation and SQL Lab 2: Advanced SQL Lab 3: NoSQL database Lab 4: Cloud database																																						
Teaching/Learning Methodology	Lectures: Fundamental principles and key concepts of the subject are delivered to students. Tutorials: Students will be able to clarify concepts and have a deeper understanding of the lecture material; problems and application examples are given and discussed. Students will be given programming exercises and use database development tools to design databases. Laboratory Sessions: Students will do some programming exercises to enhance their understanding of database design and development.																																						
Alignment of Assessment and Intended Subject Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific Assessment Methods/Tasks</th><th rowspan="2">% Weighting</th><th colspan="4">Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)</th></tr> <tr> <th>1</th><th>2</th><th>3</th><th>4</th></tr> </thead> <tbody> <tr> <td>Assignment</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Tests/quizzes</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>Lab works</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr> <tr> <td>Total</td><td>100%</td><td></td><td></td><td></td><td></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignment: Students are required to explore selected database topics and deliver video presentations to assess their communication skills. Tests/quizzes: Quizzes are used to measure the student's understanding of the theories and concepts as well as their comprehension of the subject materials. Tests are used to evaluate the student's ability to apply concepts and skills learned in the classroom; students need to think critically and learn independently in order to come up with an appropriate design. Lab works: Students are required to work on labs regarding SQL, NoSQL, and cloud databases.					Specific Assessment Methods/Tasks	% Weighting	Intended Subject Learning Outcomes to be Assessed (Please tick as appropriate)				1	2	3	4	Assignment	10%	✓	✓	✓	✓	Tests/quizzes	50%	✓	✓	✓		Lab works	40%	✓	✓	✓		Total	100%				
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Lab works	40%	✓	✓	✓																																			
Total	100%																																						
Student Study Effort Expected	Class contact (time-tabled):																																						
	• Lecture/Tutorial		27 Hours																																				
	• Laboratory/Practice Classes		12 Hours																																				
	Other student study effort:																																						
	• Lecture: preview/review of notes; homework/assignment; preparation for test/quizzes/examination		36 Hours																																				
	• Tutorial/Laboratory/Practice Classes: preview of materials, revision and/or reports writing		30 Hours																																				
	Total student study effort:		105 Hours																																				

Reading List and References	1. Thomas Connolly and Carolyn Begg, <i>Database Systems: A Practical Approach to Design, Implementation, and Management</i>, 6th ed., Pearson, 2019. 2. Mark L. Gillenson, <i>Fundamentals of database management systems</i>, Wiley, 3rd ed., Wiley, 2023. 3. Dan Sullivan Sullivan, <i>NoSQL for Mere Mortals</i>, Addison-Wesley Professional, 2015. 4. Shannon Bradshaw, Eoin Brazil, and Kristina Chodorow, <i>MongoDB: The Definitive Guide: Powerful and Scalable Data Storage</i>, 3rd ed., O'Reilly Media, 2019. 5. P. Lakhera, “Creating a Relational Database in the Cloud using AWS Relational Database Service (RDS),” in <i>AWS for System Administrators</i>, United Kingdom: Packt Publishing, Limited, 2021.
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